Agentic AI: Towards Smarter and Independent AI Systems

Agentic AI systems can think and act independently and are far more intelligent than classic AI and generative AI systems. As they evolve, they will change the world around us.

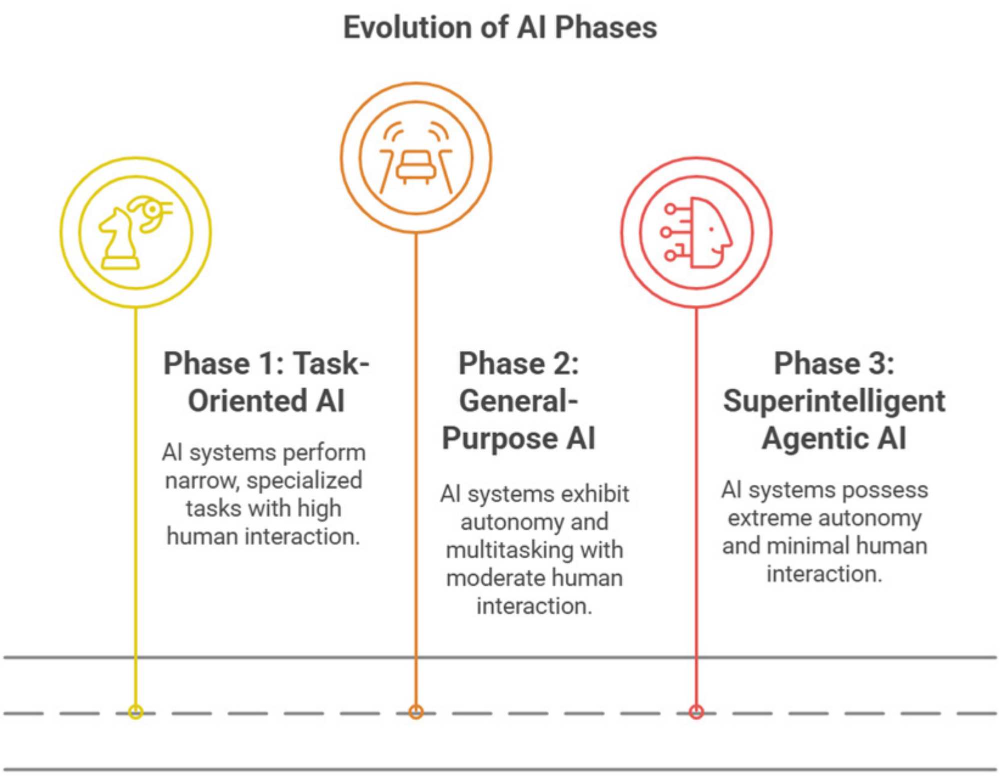

Figure 1: Evolution of AI

Artificial intelligence (AI) has evolved a great deal since the 1950s. When it was first introduced, the AI system used fixed rules. We are now moving towards generative AI, based on newer technologies such as large language models (LLMs) and natural language processing (NLP), examples of which are ChatGPT and Google Gemini.

Agentic AI is a new artificial intelligence paradigm that focuses on the design of agents that can act independently to achieve specified goals. Unlike traditional AI systems, which generally perform based on a specific set of inputs, agentic AI is designed to decide and act based on environmental cues and inherent objectives. Hence, agentic AI systems are highly capable of taking the initiative and assessing outcomes based on their experience. This independence is attained by employing sophisticated methods like reinforcement learning, hierarchical goal definition, and ongoing learning.

In agentic AI, there are systems where specified parameters, goals, and decision-making capabilities of the agent assist in addressing many scenarios, which could be at times interconnected with other agents. In that sense, agentic AI is not merely processing data but instead a system based on "self-driven" intention, which is distinct from all other AI systems. This is a significant step forward in artificial intelligence. Agentic AI has numerous implications for robotics and virtual assistants, as well as for systems in finance and healthcare, where being capable of making autonomous, real-time decisions is fundamental.

Conventional AI systems are rule-based or algorithm-dependent. They cannot diverge from their programmed directives and obey rules rigidly. Agentic AI is programmed to make choices autonomously, find solutions, and perform actions independently based on experiences acquired and objectives. What sets agentic AI apart is that it has autonomy -- the system is not watched at all times. This implies that it also does intelligent decision-making, where it looks at conditions and then